Algebra I: Exponent Rules

Timed Practice — 30 Minutes

Name: _____ Date: _____

Instructions

Complete all 18 problems in 30 minutes without a calculator. You may use this reference page. Show key steps and give exact answers. Write simplified answers with positive exponents only; do not leave fractions within fractions. Assume all variables in expressions being simplified are positive real numbers. In Problem 16, solve for real x.

Fundamental Exponent Rules

For the integer-exponent rules below, $m, n \in \mathbb{Z}$ and all expressions must be defined. In particular, denominators and bases raised to zero or negative powers must be nonzero.

Rule	**Formula**
First power	$a^1 = a$
Product of powers	$a^m \cdot a^n = a^{m+n}$
Quotient of powers	$\dfrac{a^m}{a^n} = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Power of a product	$(ab)^n = a^n b^n$
Power of a quotient	$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$
Zero exponent	$a^0 = 1 \quad (a \neq 0)$
Negative exponent	$a^{-n} = \dfrac{1}{a^n}, \qquad \left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$
Rational exponent	$a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$

For the rational-exponent rule, take $a > 0$, $m \in \mathbb{Z}$, and positive integer n. The product, quotient, and power rules also hold for rational exponents when all bases are positive.

Important Conditions and Conventions

- For positive integers n, $0^n = 0$. The expressions 0^0 and 0^{-n} are undefined here.
- A negative exponent means a reciprocal, not a negative value.
- For positive integers n, $(-a)^n = a^n$ when n is even and $(-a)^n = -a^n$ when n is odd. In $-a^n$, the exponent applies to a only.
- Exponents distribute over products and quotients, not sums or differences. In general, $(a+b)^n \neq a^n + b^n$.

Exponent Rules Practice

Problems 1–6: approximately 6 minutes

Simplify each expression.

1. $x^7 \cdot x^4$

2. $\dfrac{y^{12}}{y^5}$

3. $(a^3)^4$

4. $(-3x^2y)^3$

5. $\left(\dfrac{4a^3}{3b^2}\right)^2$

6. $(7x^3y^2)^0 + 2^0$

Exponent Rules Practice — Continued

Problems 7–12: approximately 10 minutes

Simplify each expression.

7. $\left(\frac{5}{2}\right)^{-2}$

8. $-2^4 + (-2)^4 + (-2)^3$

9. $\frac{12x^{-3}y^5}{3x^2y^{-1}}$

10. $(2a^{-2}b^3)^{-2}$

11. $\frac{(-3m^2n^{-1})^2(2m^{-3}n^4)}{6m^{-1}n^2}$

12. $\left(\frac{4x^3y^{-2}}{2x^{-1}y}\right)^{-2}$

Exponent Rules Practice — Continued

Problems 13–18: approximately 14 minutes

13. Evaluate exactly: $16^{3/4} + 27^{2/3}$.

14. Write $x^{5/2}$ in simplified radical form, and write $\sqrt[3]{a^4}$ using a rational exponent.

15. Simplify: $\dfrac{(x^{3/2}y^{-1/3})^2}{x^{-1}y^{1/3}}$.

16. Solve: $3^{2x-1} = \dfrac{27^x}{9}$.

17. A student writes $(2a^2b^{-1})^3 = \dfrac{6a^5}{b^3}$. Explain each error and give the correct simplification.

18. Simplify $\dfrac{x^{-2} + y^{-2}}{x^{-1}y^{-1}}$, then evaluate your result at $x = 2$, $y = 3$.